

# Water Electrolysis

## Experiment Packet

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**Audience:** instructors and teaching assistants

**Purpose:** Bench instructions for instructors and TAs

**Session hook:** Split water and see a 2:1 ratio

**Length:** 90-minute lab block (~20 min concepts + ~70 min hands-on)

### This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

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Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

## Session 4 — Water Electrolysis and Gas Collection

**Hook:** Split water, collect gases, and see a 2:1 stoichiometric ratio.

### Learning outcomes

- Build a gas collection apparatus by water displacement
- Run water electrolysis safely with a non-chloride electrolyte
- Measure H<sub>2</sub> and O<sub>2</sub> volumes and compare to 2:1 ratio
- Connect electrolysis to hydrogen as an energy storage concept
- **Start** electrolytic rust-removal cells for Session 5 reveal

### Session plan (90 min)

| Time      | Activity                                                      | Reference              |
|-----------|---------------------------------------------------------------|------------------------|
| 0–10 min  | Hook: inverted water-filled tubes — which gas appears faster? | lecture.md             |
| 10–25 min | Water splitting and stoichiometry                             | lecture.md             |
| 25–40 min | Build gas collection setup                                    | experiment.md — Part A |
| 40–60 min | Run electrolysis; record volumes over time                    | experiment.md — Part B |
| 60–70 min | Discuss ratio; optional instructor H <sub>2</sub> pop test    | experiment.md — Part C |
| 70–80 min | Brief energy storage discussion                               | lecture.md             |
| 80–90 min | <b>Start derusting cells</b> for Session 5                    | experiment.md — Part E |

### Key reactions

Overall:  $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$

Cathode (–):  $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$

Anode (+):  $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$  (acidic media)

or  $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$  (basic media)

### Status

- [ ] Graphite electrodes sourced
- [ ] Gas collection apparatus pilot-tested
- [ ] **No saltwater** — baking soda or Na<sub>2</sub>SO<sub>4</sub> only
- [ ] Pop test protocol reviewed
- [ ] Rusty nails + washing soda ready for Part E

## Session 4 — Preparation

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### Before class (instructor)

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- ☐ **Pilot full setup** — confirm 2:1 ratio visible within session time
- ☐ Pre-cut graphite electrodes to useful lengths
- ☐ Mix electrolyte batch: water + baking soda (label "NO SALT")
- ☐ Test tubes: practice inversion technique — no trapped air
- ☐ Review pop test safety; decide if demo will run (weather, room rules)
- ☐ Optional: test solar panel output with same cell
- ☐ Stage rusty nails, washing soda, sacrificial anodes, and spare jars for **Part E** (end of class)
- ☐ Confirm overnight power policy for derusting cells (prep room vs. disconnect)

### Day-of setup

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- ☐ Pre-fill tub at each station (or central fill station)
- ☐ Pre-fill inverted tubes underwater as demo for students
- ☐ Large sign: "**No salt in electrolysis**"
- ☐ Separate instructor area for pop test marked off
- ☐ Part E station: rusty nails, washing soda, steel/graphite anodes, labels, camera

### Common failure modes (from pilot)

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| Issue                           | Prevention                        |
|---------------------------------|-----------------------------------|
| Air in tubes                    | Demo inversion step slowly        |
| Weak bubble rate                | Fresh baking soda; clean graphite |
| Students add salt "to speed up" | Pre-brief; labeled solutions      |

### Open questions / notes

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- Electrode material used:
- Best tube type for measurement:
- Pop test yes/no for this venue:

## Session 4 — Materials

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### Per group

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- ☐ Clear plastic tub or container
- ☐ Water (~500 mL per tub)
- ☐ Baking soda: **1–2 tsp per 500 mL** (~0.1–0.2 M) **or** sodium sulfate ~0.1 M (~7 g / 500 mL)
- ☐ Graphite electrodes (thick pencil leads or carbon rods)
- ☐ 9 V battery or DC power supply
- ☐ Alligator clips
- ☐ Two small test tubes, vials, syringes, or narrow plastic bottles
- ☐ Ruler (volume estimation)
- ☐ Goggles

### Optional

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- ☐ pH indicator
- ☐ Small solar panel
- ☐ Graduated cylinders for calibration

### Instructor-only (pop test)

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- ☐ Small flame source (long lighter)
- ☐ Separate tiny collection vessel
- ☐ Fire extinguisher accessible

### Part E — Start rust removal (end of class)

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- ☐ Rusty iron nails or hardware (2+ per group)
- ☐ Washing soda: **1 tbsp per ~250 mL** warm water (preferred) **or** baking soda same measure (NOT NaCl)
- ☐ Sacrificial steel strips or graphite rods
- ☐ Small jars or cups
- ☐ Labels and camera for before photos

### Explicitly excluded

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- ☐ ~~Table salt (NaCl)~~ — **do not use**

### Pilot notes

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| Parameter   | Value from pilot                              |
|-------------|-----------------------------------------------|
| Electrolyte | Baking soda / Na <sub>2</sub> SO <sub>4</sub> |

| Parameter                      | Value from pilot |
|--------------------------------|------------------|
| Time to visible gas difference | ___ min          |
| Typical $V_{H_2}/V_{O_2}$      | ___              |

## Session 4 — Experiment: Water Electrolysis

**Instructors:** gas ID, ratio systematics, Part E polarity — instructor.md.

### Electrolyte recipes — concentrations

**Never use table salt (NaCl)** — chloride can produce toxic chlorine gas at the anode.

#### Option A — Baking soda (recommended, easy)

| Batch             | Baking soda ( $\text{NaHCO}_3$ ) | Water   |
|-------------------|----------------------------------|---------|
| ~500 mL (one tub) | <b>1–2 tsp</b> (~5–10 g)         | ~500 mL |
| ~1 L              | <b>2–4 tsp</b> (~10–20 g)        | ~1 L    |

Stir until dissolved. Label **"baking soda electrolyte — NO SALT"**.

Approximate concentration: **~0.1–0.2 M  $\text{NaHCO}_3$**  (kitchen measuring is fine; exact molarity is not critical).

**Water:** tap water is fine — the baking soda (or  $\text{Na}_2\text{SO}_4$ ) provides the ions you need. Distilled water is unnecessary here.

#### Option B — Sodium sulfate (optional)

| Batch  | $\text{Na}_2\text{SO}_4$ (anhydrous) | Water  |
|--------|--------------------------------------|--------|
| 500 mL | <b>~7 g</b>                          | 500 mL |

≈ **0.1 M  $\text{Na}_2\text{SO}_4$** . Clear, effective supporting electrolyte.

#### Volume per station

Fill the tub deep enough to fully cover inverted tube mouths (~3–5 cm liquid depth typical). Mix **fresh** electrolyte if someone contaminated a tub with salt.

## Part A — Build gas collection setup (25–40 min)

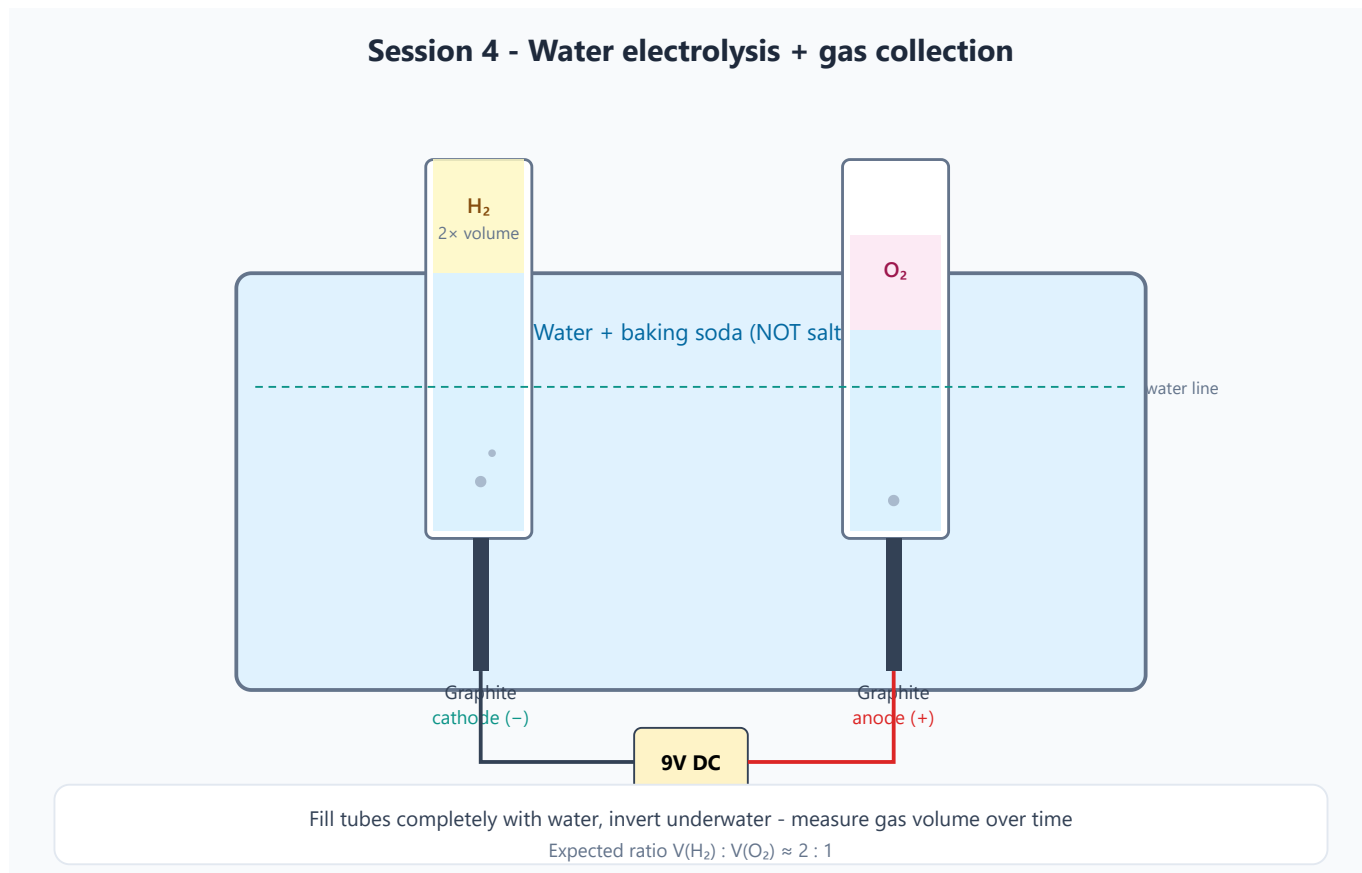


Figure 1 — Inverted water-filled tubes collect  $\text{H}_2$  (2× volume) and  $\text{O}_2$ . Use baking soda or sodium sulfate — not salt.

### Materials

- Clear tub + electrolyte (Option A or B)
- Two small test tubes, vials, or narrow bottles (same size preferred)
- Two graphite electrodes (pencil leads ~2–3 mm or carbon rods)
- 9 V battery + alligator clips
- Ruler or graduated markings for gas column height
- Goggles

### Build steps

1. Fill the tub with electrolyte.
2. Fill tube 1 **completely** with the same electrolyte — no air bubble at the closed end.
3. Cover the mouth with a finger or card; invert; lower into the tub; release underwater. Tube stays full of liquid.
4. Repeat for tube 2.
5. Slide a graphite electrode up into each tube (or position electrodes so bubbles rise into each tube).
6. Connect:

7. **Cathode (–)** / black → electrode in tube that will collect **H<sub>2</sub>** (will fill faster)
8. **Anode (+)** / red → other electrode (**O<sub>2</sub>**)
9. Keep battery clips **dry** (not submerged).
10. When current flows, the **faster-filling** tube is hydrogen (cathode).

#### Pre-run checklist

- [ ] Both tubes liquid-filled with **no air gap** at start
  - [ ] Electrodes secure; battery clips dry
  - [ ] Electrolyte is baking soda or Na<sub>2</sub>SO<sub>4</sub> — **not salt**
  - [ ] Goggles on
  - [ ] Tubes labeled H<sub>2</sub> / O<sub>2</sub> after first bubbles appear
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#### Part B — Run and measure (40–65 min)

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1. Start the timer when bubbles are clearly visible at both electrodes.
2. Every **2–3 min**, measure gas volume in each tube:
3. Prefer graduated tubes (mL), **or**
4. Measure gas column height (cm) with a ruler (ratio of heights ≈ ratio of volumes if tubes match).
5. Continue until a clear difference appears or ~15–20 min have elapsed.

#### Gas volume log

| Time (min) | V_H <sub>2</sub> (mL or cm) | V_O <sub>2</sub> (mL or cm) | Ratio V_H <sub>2</sub> /V_O <sub>2</sub> |
|------------|-----------------------------|-----------------------------|------------------------------------------|
| 0          | 0                           | 0                           | —                                        |
| 3          |                             |                             |                                          |
| 6          |                             |                             |                                          |
| 9          |                             |                             |                                          |
| 12         |                             |                             |                                          |
| 15         |                             |                             |                                          |
| 20         |                             |                             |                                          |

#### Expected result

V\_H<sub>2</sub> / V\_O<sub>2</sub> ≈ **2** (ideal). Classroom range **1.6–2.2** is a success — discuss solubility and collection errors.

#### Observations

- Bubble rate at each electrode
  - Cloudiness or electrode crumbling
  - Temperature change (optional)
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## Part C — Instructor-only pop test (65–75 min)

### Optional demo — not for students

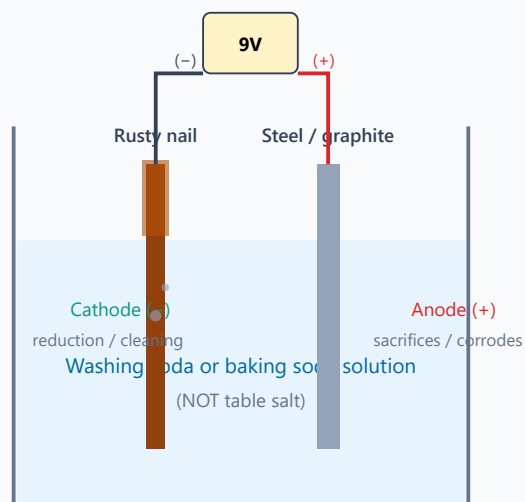
1. Collect a **tiny** amount of  $\text{H}_2$  in a separate small tube.
2. Move away from the main tub and faces.
3. Brief flame → soft “pop.”
4. Never ignite sealed or mixed  $\text{H}_2/\text{O}_2$  containers.
5. Skip if the venue forbids open flame.

## Part D — Extensions (if time)

- [ ] Add universal indicator — look for acid/base zones near electrodes
- [ ] Power with a small solar panel instead of a battery
- [ ] Compare baking soda vs. 0.1 M sodium sulfate side by side

## Part E — Start electrolytic rust removal (80–90 min)

### Electrolytic rust removal (start Session 4, reveal Session 5)



Start at end of Session 4; rinse and compare to control nail at start of Session 5  
Same electrolysis idea as Session 4 - electricity drives a non-spontaneous reaction

Figure 2 — Start cells now; reveal cleaned nails at opening of Session 5.

### Derusting electrolyte

| Ingredient                                                           | Amount                                |
|----------------------------------------------------------------------|---------------------------------------|
| Washing soda (sodium carbonate, $\text{Na}_2\text{CO}_3$ ) preferred | 1 tbsp (~15 g) per ~250 mL warm water |
| or baking soda                                                       | 1 tbsp per ~250 mL warm water         |

≈ **0.5–0.6 M** Na<sub>2</sub>CO<sub>3</sub> if using washing soda at that kitchen measure — exact molarity is not critical.

**Do NOT use NaCl.**

### Setup steps

1. Dissolve washing soda (or baking soda) in warm water; pour into a small jar (~100–250 mL).
2. Clip **rusty iron** nail/hardware to battery **cathode (–)**.
3. Clip **sacrificial steel strip or graphite** to battery **anode (+)**.
4. Immerse both; keep them from touching; confirm gentle bubbling.
5. Label group name; photograph the nail **before**.
6. Place a **control** rusty nail in plain water (no battery).
7. Leave powered overnight per instructor / venue rules.

### End-of-class checklist

- ☐ All derusting cells connected and bubbling (or current confirmed)
- ☐ Control nails in plain water labeled
- ☐ Before photos taken
- ☐ Overnight power policy confirmed (prep room preferred)

### Troubleshooting

| Problem               | Cause                           | Fix                                 |
|-----------------------|---------------------------------|-------------------------------------|
| No bubbles            | Poor contact / weak electrolyte | Check clips; add more baking soda   |
| One side only         | Single electrode connected      | Check both leads                    |
| Ratio far from 2:1    | Air in tube at start; leaks     | Rebuild tubes with no air gap       |
| Chlorine smell        | <b>Salt used</b>                | Stop; dump; remake baking soda bath |
| Tubes won't stay full | Air introduced on invert        | Practice underwater release         |

### Safety checklist

- ☐ No NaCl electrolyte at this station
- ☐ No flames near the collection setup (except instructor pop test)
- ☐ Small gas volumes only
- ☐ Pop test instructor-only, at a safe distance

### Experiment status

- ☐ Electrolyte recipe posted (baking soda amounts)
- ☐ Tube sizing / volume method chosen
- ☐ H<sub>2</sub> vs. O<sub>2</sub> side identified in pilot (cathode = H<sub>2</sub>)

- [ ] Typical ratio from pilot: \_\_\_\_
- [ ] Rust-removal jars + washing soda staged for Part E

## Syllabus Safety Notes

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Apply these across all sessions. Session-specific hazards are listed in each session folder.

### General rules

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1. **No eating experimental materials.** Fruits, potatoes, and liquids used in experiments are no longer food.
2. **Avoid short circuits.** Batteries can heat up if directly shorted.
3. **Clean surfaces matter.** Electroplating requires clean metal surfaces.
4. **Metal salt waste should be collected.** Copper and silver solutions should not be casually poured down the drain.
5. **Water choice:** tap water is fine for most baths; use **distilled/deionized water for 0.01 M AgNO<sub>3</sub>** (Session 3) so chloride does not precipitate silver.

### Session-specific hazards

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| Session | Key hazard                               | Mitigation                                                                            |
|---------|------------------------------------------|---------------------------------------------------------------------------------------|
| 1       | Sharp nails/wires                        | Supervise handling; wash hands after metals                                           |
| 2       | Copper solutions                         | Goggles; no ingestion; collect waste                                                  |
| 3       | Silver nitrate                           | Gloves + goggles; stains skin/clothing; small volumes                                 |
| 4       | Gas evolution                            | No saltwater; no flames near setup; instructor-only pop test                          |
| 5       | Corrosion demos + electrolytic derusting | Use washing soda or baking soda only — NOT NaCl; disconnect power when not supervised |

### Hydrogen pop test (Session 4, instructor-only)

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- Use a **tiny** volume of hydrogen only
- Perform away from the electrolysis setup
- **Never** ignite a sealed container or a mixed H<sub>2</sub>/O<sub>2</sub> container